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Education

- **Ph.D. in Physical Chemistry**, Beijing Normal University, 2022
Advisors: Jiu-Shu Shao (邵久书), Jian Liu (刘剑)
- **B.S. in Chemistry**, Boling College (化学伯苓班), Nankai University, 2015
Advisors: Peng Cheng (程鹏), Wei Shi (师唯), Hong-Wei Sun (孙宏伟)

Research Experience

- **Postdoc**, School of Physics, Peking University, 2023–
Advisor: Xin-Zheng Li (李新征)
- **Research Assistant**, Peking University, 2022
Advisor: Jian Liu (刘剑)

Publications

1. **Wang**[†], He[†], Zeng, Yang, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. III. Re-weighted simulation and identification of zeros.
2. He[†], Zeng[†], Yang, **Wang***, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. II. Finite temperature results and continuation on the complex plane, arXiv: 2606.07415.
3. He[†], Zeng[†], Yang, **Wang***, Ye*, Li*, Revisiting the fermion sign problem from the structure of Lee-Yang zeros. I. The form of the partition function for anyons and its zeros at 0 K, Phys. Rev. E, **113**, 024115 (2026).
4. Liang[†], Wang[†], **Wang**[†], Wang, She, Wang, Shao, Liu*, An efficient integrator scheme for sampling the (quantum) isobaric–isothermal ensemble in (path integral) molecular dynamics simulations, J. Chem. Theory Comput. **21**, 6394 (2025).
5. **Wang**[†], Zhang[†], Liu*, Shao*, Generalized fourth-order decompositions of imaginary time path integral: Implications of the harmonic oscillator, Chin. J. Chem. Phys. **35**, 516 (2022).
6. Li, Han, Chai, **Wang**, Zhang, Chen, Liu*, Shao*, Stationary state distribution and efficiency analysis of the Langevin equation via real or virtual dynamics, J. Chem. Phys. **147**, 184104 (2017).
7. **Wang**, Zhang, Liu*, Shao*, An efficient algorithm for the fourth-order decomposition for path integral molecular dynamics.
(In preparation)
8. **Wang**, Zhang, Liu*, Shao*, Application of the generalized fourth-order decomposition schemes to perturbed path integral.
(In preparation)

Talks and Presentations

- Revisiting the fermion sign problem from the structure of Lee-Yang zeros, The 19th National Conference on Chemical Dynamics, September 18–21, 2025, Chun'an, Hangzhou, Zhejiang
(Oral report)

- Fermion sign problem and distribution of Lee-Yang zeros, Frontiers in Ab Initio Quantum Molecular Dynamics (CECAM Flagship School), Jul 25–29, 2026, Beijing (Awarded poster)
- Lee-Yang zeros and path integral molecular dynamics of indistinguishable particles, The 35th Chinese Chemical Society Congress, Apr 11–14, 2026, Chongqing (Awarded poster)
- Efficient algorithms for sampling canonical and isobaric ensembles via molecular dynamics, The 31st Chinese Chemical Society Congress, May 5–8, 2018, Hangzhou, Zhejiang (Awarded poster)

Honors and Awards

- Functional Molecular Materials Research Group of Nankai University, Deng Xiaoping’s Scientific and Technological Innovation Team (小平科技创新团队), China Youth Science and Technology Innovation Award Fund, 2014
Team leader

Research Interests

- **Acceleration of path integral molecular dynamics (PIMD) methods**

Path integral molecular dynamics (PIMD) is an effective method for studying nuclear quantum effects in chemical and physical systems, but its enormous computational cost limits its potential applications to biological and solvated systems.

- We developed the “middle scheme” thermostat algorithm, which enlarges the sampling time step by adjusting the order of the thermostat coupling, and applied it to the sampling of the canonical ensemble (JCP 2017) and the isobaric-isothermal ensemble (JCTC 2025) in classical and path integral molecular dynamics. **This algorithm has been widely adopted by software packages such as Amber.**
- We developed parameterized generalized fourth-order decompositions (factorized generalized fourth-order decomposition) to replace the second-order Trotter splitting, and optimized the parameters through analysis of the analytically solvable one-dimensional harmonic oscillator (CJCP 2022). **This work has been cited multiple times by Prof. Chin of Texas A&M University, a senior researcher in path integral methods and the originator of fourth-order decomposition algorithms.** The fourth-order decomposition methods and optimized parameters can be applied to finite-difference-based open-chain and closed-chain PIMD, effectively improving the sampling efficiency and enlarging the accessible systems (to be published).
- We proved that the perturbed path integral (PPI) is at least fourth-order accurate and can be combined with generalized fourth-order decompositions to further improve the computational efficiency of PIMD by tuning the parameters (to be submitted).

- **Fermion sign problem and PIMD of indistinguishable particles**

In systems involving exchange symmetry, PIMD for indistinguishable particles (bosons with $\xi = 1$, fermions with $\xi = -1$) is an effective method for studying superfluid and supersolid systems. At present, the computational cost of bosonic PIMD has been reduced to scale quadratically with system size, but potential sampling ergodicity problems remain; fermionic PIMD is limited by the sign problem (FSP) arising from the cancellation of positive and negative contributions under particle exchange, making it difficult to apply to large systems and at low temperatures.

- We connected the FSP with the Lee-Yang theory of phase transitions, extended ξ to the complex plane, and discovered that the partition function is a polynomial in ξ , so that all systems possess fixed zeros at zero temperature in the ξ space that depend only on the particle number (PRE 2026). **This work has been cited multiple times by Dornheim, a leading researcher**

in warm dense matter and head of the High Energy Density Division at Helmholtz-Zentrum Dresden-Rossendorf, Germany.

- We used Lee-Yang zeros to explain the origin of the FSP, analyzed the movement of the zeros at finite temperature using analytically solvable non-interacting systems, and proposed a method to obtain the low-temperature properties—which are severely affected by the FSP—by analytic continuation from the high-temperature thermodynamic properties of fermionic systems (submitted to PRE).
- We proposed a method to obtain the Lee-Yang zeros of general systems by numerical computation, studied the influence of the zeros on the physical properties of interacting systems, and used the zero distribution to guide the calculation of thermodynamic properties in fermionic PIMD (to be published).

These three papers form a series that establishes a theoretical framework for studying the FSP via Lee-Yang theory, explains the limitations of current path-integral-based fermionic methods, and guides the development of methods for solving the fermion sign problem.

Future Research Plans

- **Acceleration methods for path integrals of indistinguishable particles**

Path integral methods for indistinguishable particles have been applied to the exploration of super-solids and to simulations of systems such as the uniform electron gas (UEG), warm dense hydrogen, and beryllium, where the results can be compared with X-ray Thomson scattering data from the US National Ignition Facility (NIF). However, these methods remain limited by the growth of computational cost with system size and by the FSP. These problems can be alleviated by accelerating the path integral.

- The normal-mode and staging coordinate transformations available for distinguishable particles cannot decouple the coordinates of indistinguishable particles, which constrains the sampling efficiency. We plan to develop new coordinate transformations to improve the sampling efficiency.
- Studies have shown that combining fourth-order path integral methods with approaches such as permutation blocking and higher-order fermionic propagators can alleviate the FSP. Our tunable-parameter generalized fourth-order decompositions can be applied to both bosonic and fermionic systems; combined with machine-learning-based parameter tuning for different systems, they are expected to further expand the system sizes and temperature ranges accessible to path integral methods.

- **Real-time dynamics methods for indistinguishable particles**

At present, PIMC for distinguishable particles exhibits higher sampling efficiency than PIMD and is therefore more widely used for sampling thermodynamic properties, but real-time dynamical properties cannot be obtained from PIMC. Approximate dynamics methods based on PIMD for distinguishable particles, such as ring polymer molecular dynamics (RPMD) and centroid molecular dynamics (CMD), have theoretical limitations but strong transferability across systems, and have been widely applied to spectral calculations and reaction rate calculations. Explorations of RPMD methods for bosons have already begun. Introducing exchange symmetry into dynamics studies is expected to enable accurate descriptions of the dynamics of reactions such as hydrogen and electron transfer, as well as spectroscopic problems such as tunneling splittings.

- Calculate the dynamic structure factor of the UEG using real-time dynamics methods, replacing the approach currently used by Dornheim and others—reconstructing the dynamic structure factor by analytic continuation of PIMC imaginary-time correlation functions—thereby overcoming the latter’s inherent mathematical ill-posedness.
- Combine existing fermionic path integral methods with RPMD and apply them to the study of tunneling splittings of methyl rotations in polymethyl systems; the results may find potential applications in areas such as selectivity in organic chemical reactions, coarse-grained simulations of macromolecules, and combustion chemistry.

- Combine fermionic path integral methods with dissipative dynamics methods and apply them to reactions involving nuclear quantum effects and exchange symmetry on surfaces and in cages (e.g., $\text{H} + \text{H}_2$ and more complex reactions); the results may address important questions in astrochemistry such as reactions on interstellar ice surfaces.

The reaction systems above involve a huge number of vibrational modes that are difficult to treat with existing wave-function methods (such as multiconfiguration time-dependent Hartree, MCTDH), but fermionic-path-integral-based dynamics methods are expected to enable the study of the role of exchange symmetry in these systems.

Referees

- Xin-Zheng Li (李新征) School of Physics, Peking University zxli@pku.edu.cn
- Jiu-Shu Shao (邵久书) College of Chemistry, Beijing Normal University jiushu@bnu.edu.cn
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